FINAL PROJECT · DEEP LEARNING & NLP

Contract Clause Review Assistant

Classifying, retrieving and auditing contract clauses with BERT

Legal NLP | BERT · LEDGAR

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THE BRIEF

A review assistant for contract clauses

It reads a clause, says what it is, finds similar clauses and knows its limits.

MISSION 1

Classify

Assign each clause one of 100 types with a fine-tuned BERT

MISSION 2

Pipeline

Tokenize, train and evaluate on a held-out test set

MISSION 3

Retrieve

Find clauses by meaning, not keywords, with a Sentence-Transformer

MISSION 4

Audit

Biases, errors, limits and uses to avoid

THE DATA

LEDGAR: 80,000 real contract clauses

80,000

clauses from public US contracts filed with the SEC

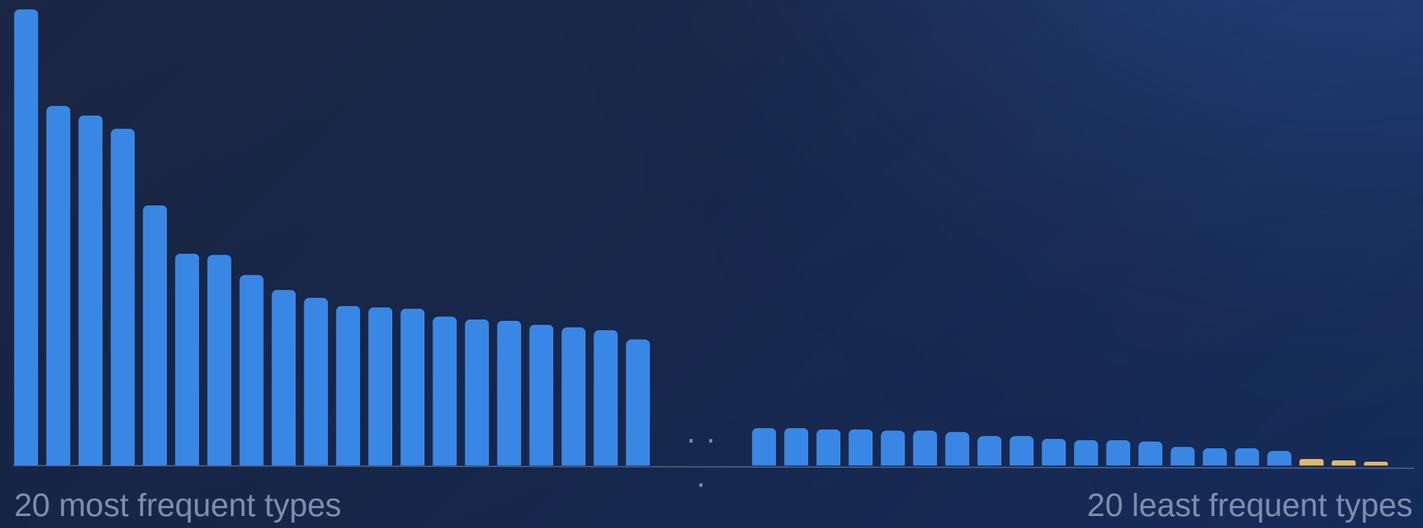
100

clause types · split 60k / 10k / 10k

140 : 1

imbalance between the most and least frequent type

Training examples per type: the 20 most and 20 least frequent



Gold: fewer than 50 examples (Qualifications, Assigns, Books). This is why we track macro-F1, not only accuracy.

THE METHOD

One pipeline, two uses

CLASSIFICATION · MISSIONS 1 AND 2

Clause
Raw contract text, in English

→

Tokens
WordPiece, at most 256 tokens

→

BERT
Fine-tuned 2 epochs on a T4 GPU

→

Label
One of the 100 clause types

RETRIEVAL · MISSION 3

Query
A question or a pasted clause

→

Encode
all-MiniLM-L6-v2, 384 dimensions

→

Index
60,000 training clauses, no test clause

→

Top-k
Closest clauses, with a score

⚠ **Audit (mission 4)**: per-type errors, long clauses, confidence, limits of use

RESULTS · 10,000 UNSEEN TEST CLAUSES

BERT v1 gets 84.5% of clauses right

84.5%

Accuracy

Share of clauses whose predicted type is correct

72.1%

Macro-F1

Average over all 100 types, rare ones included: our weak spot

5.8%

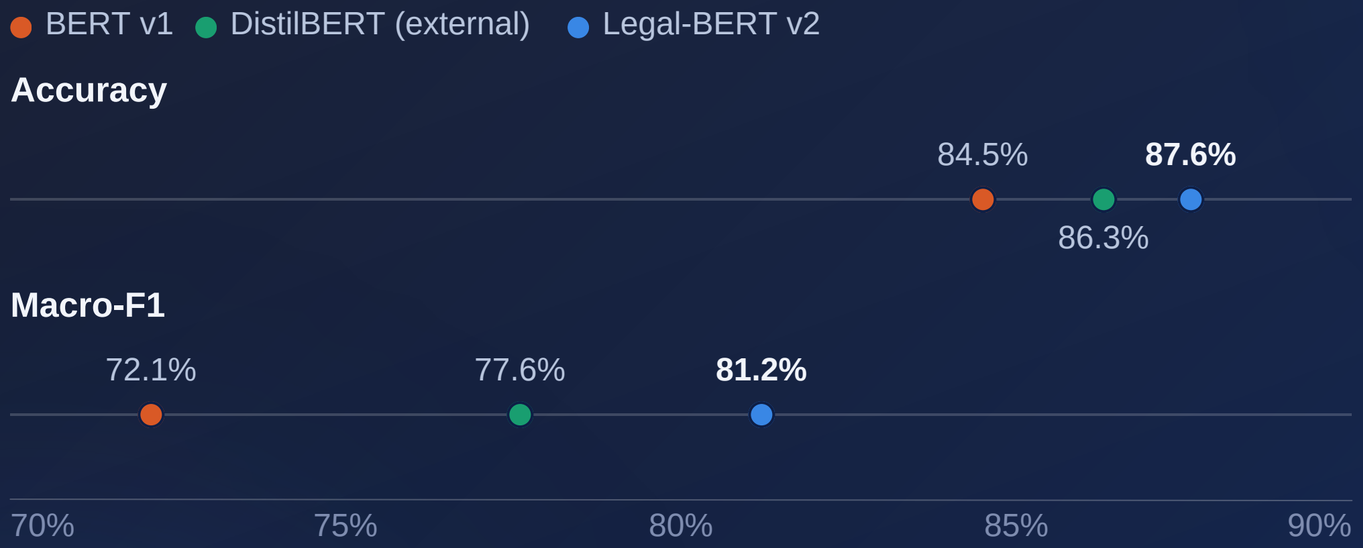
Baseline

Always predicting the most frequent type

Very reliable on standard clauses (F1): Counterparts 0.99 · Headings 0.98 · Severability 0.97 · Governing Laws 0.93

MODEL COMPARISON

Three models, one test set



TYPES WITH F1 = 0

BERT v1	9
DistilBERT	2
Legal-BERT v2	1

Rule set in advance: keep v2 only if it beats v1 on **validation**. It did: macro-F1 80.6% vs 73.2%.

McNemar test: v2 beats v1 ($\chi^2 = 112.5$) and DistilBERT ($\chi^2 = 25.1$), both $p < 0.001$.

MISSION 3 · SEMANTIC SEARCH

Finding a clause by its meaning

termination clause with 30-day notice period

Terminations — **score 0.76**

“The Term of this Agreement may be terminated by either party upon thirty (30) days’ prior written notice.”

Terminations — **score 0.74**

“You or the Company may terminate this Agreement without cause upon 30 days’ written notice to the other party…”

Notices — **score 0.73**

“Notwithstanding the foregoing, it shall be a condition precedent to the Company’s right to terminate…”

PASTED CLAUSE

“governed by the laws of the State of New York”

5 / 5

results labelled Governing Laws, scores 0.91 to 0.96

⚠ The 3rd result is about termination but labelled Notices: close in meaning is not the same legal category.

AUDIT 1 · WHERE BERT v1 FAILS

9 types at F1 = 0, for two reasons

Near-synonyms

Same vocabulary, blurry legal boundary: the model picks the more frequent “twin” type.

true type → predicted · cases			
Tax Withholdings	→	Withholdings	**72 / 75**
Applicable Laws	→	Governing Laws	**42 / 53**
Indemnity	→	Indemnifications	**27 / 29**

Too rare

Fewer than 50 training examples: not enough to learn the nuance.

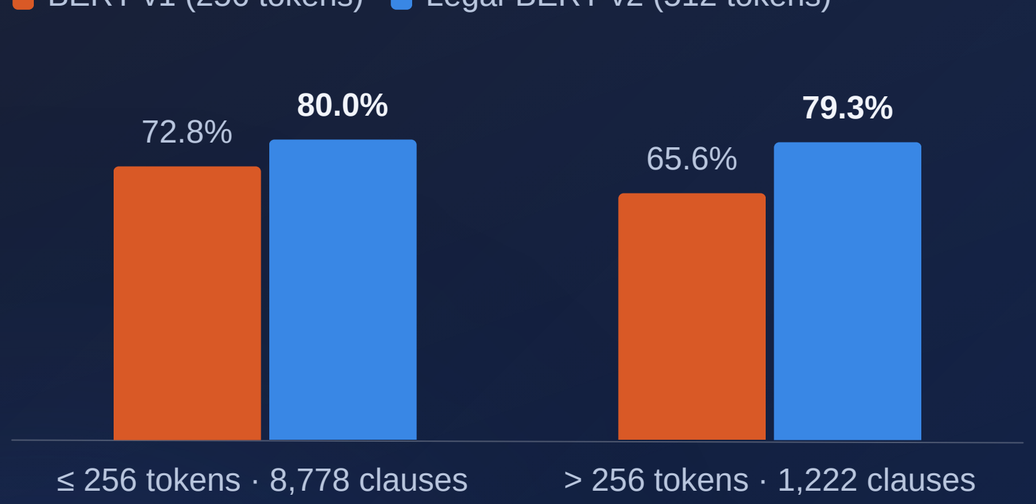
true type (examples) → predicted · cases			
Books (23)	→	Records	**2 / 2**
Assigns (31)	→	Binding Effects	**2 / 4**
Qualifications (47)	→	Organizations	**2 / 5**

The errors follow the label structure, not a pipeline bug. Version 2 recovers 8 of these 9 types (slide 12).

AUDIT 2 · LONG CLAUSES AND CONFIDENCE

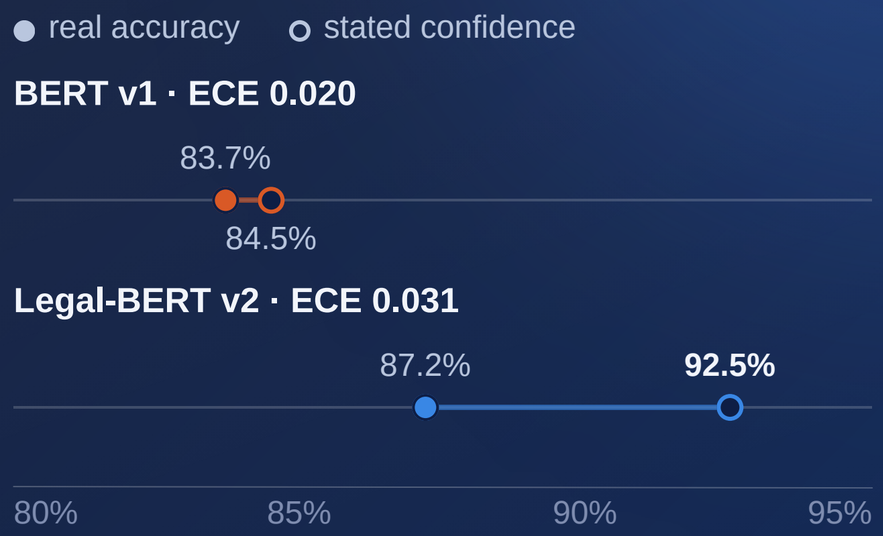
Long clauses and confidence

Macro-F1 by clause length



v1 loses **7.2 pts** on long clauses; with 512 tokens, v2 loses only **0.7 pt**.

Stated confidence vs real accuracy



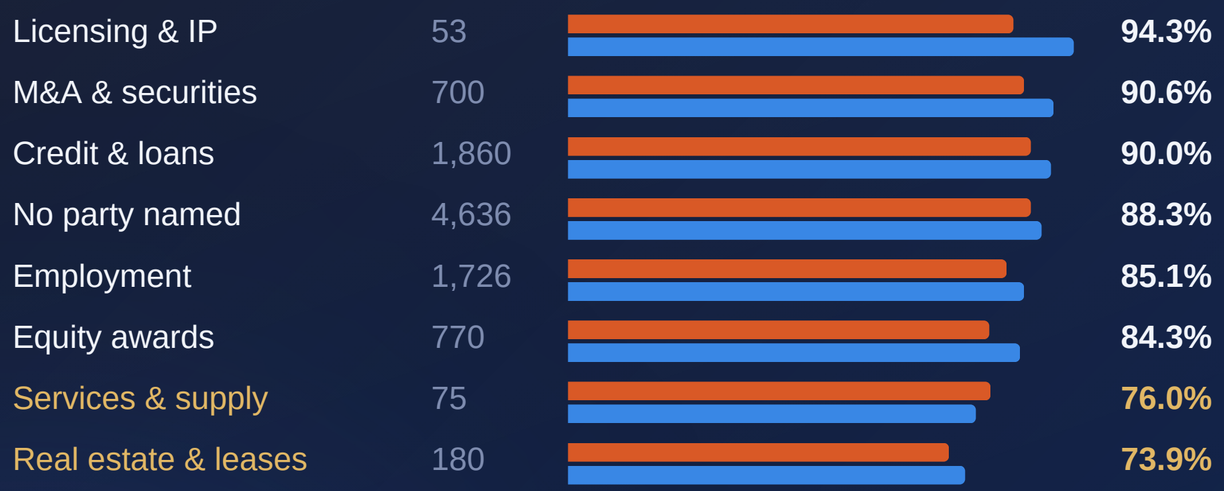
v1 is well calibrated. v2 is **overconfident by 5 pts**: temperature scaling is the next step.

AUDIT 3 · CONTRACT KINDS, PARTIES, WORDING

Does it hold across contracts and parties?

Test accuracy by contract family

BERT v1 · Legal-BERT v2 · n = test clauses



Bars start at 0%. Family inferred from the parties named in each clause.

Wording: no weak spot

v2 accuracy, with vs without

ALL CAPS clauses	**88.6% vs 87.6%**
Numbers	**89.4% vs 86.6%**
Negations	**88.5% vs 86.4%**

Not measurable here

LEDGAR only holds SEC filings of US listed companies: no consumer, SME or private contracts, no other law or language.

USE AND LIMITS

What it must not be used for

Deciding alone

No clause accepted, rejected or signed on the model’s word: about 12% are misclassified.

Weak clause types

Assigns (F1 0) and Applicable Laws (F1 0.10) are always checked by hand.

Other languages, other law

Trained only on English, US-law contracts: not for French or EU contracts.

Unfamiliar contracts

Consumer, SME or private contracts were never seen; leases and services already score 74–76%.

Judging validity

Search finds similar wording, not compliant or enforceable clauses.

Trusting the score

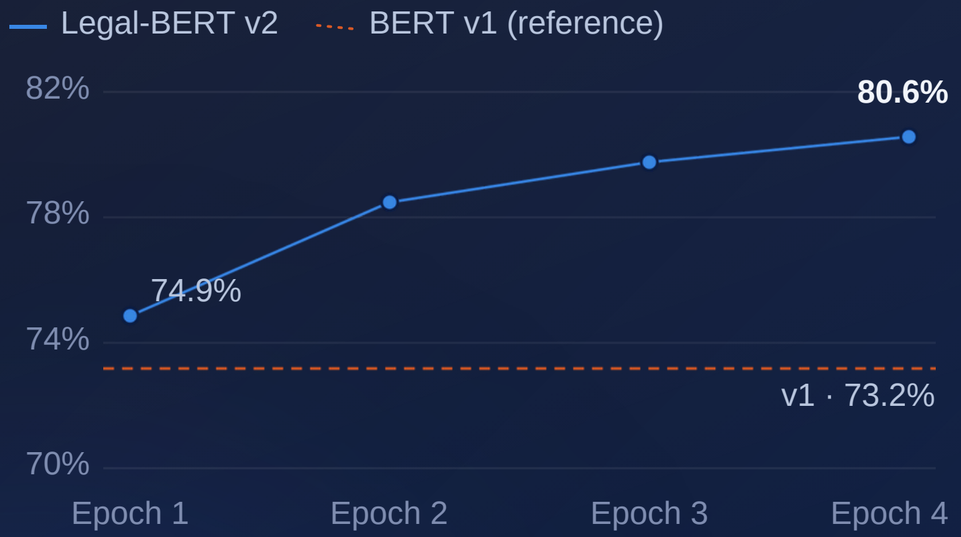
v2 overstates its confidence by about 5 pts: recalibrate before routing cases.

The model suggests; the lawyer decides.

VERSION 2 · WHAT THE AUDIT BOUGHT US

Legal-BERT, weighted loss, 512 tokens

Validation macro-F1 per epoch



Test F1 in v2 for the 9 types stuck at 0 in v1



Test set: accuracy 84.5% → **87.6%** · macro-F1 72.1% → **81.2%** · types at F1 = 0: 9 → **1** · McNemar p < 0.001

CONTRACT CLAUSE REVIEW ASSISTANT

Thank you

Questions?

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Deep Learning for Sequential Data & NLP · Albert School × Mines Paris-PSL

